Multivariate Regression Model

School of Management and Economics
The Chinese University of Hong Kong, Shenzhen



The expected value of the OLS Estimators

- **Omitting relevant variables: the simple case**

  Leaving a variable out when it should be included in multiple regression is a serious problem. This is called **excluding a relevant variable** or **underspecifying the model**.

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + u$$

True model (contains x_1 and x_2)

suppose we leave out x_2 and use simple regression of y on x_1 :

$$\tilde{y} = \tilde{\beta}_0 + \tilde{\beta}_1 x_1$$

Estimated regression fucntion (x_2 is omitted)

The expected value of the OLS Estimators

- Omitted variable bias

$$x_2 = \delta_0 + \delta_1 x_1 + v$$

If x_1 and x_2 are correlated, assume a linear regression relationship between them

$$\Rightarrow \quad y = \beta_0 + \beta_1 x_1 + \beta_2(\delta_0 + \delta_1 x_1 + v) + u$$

$$= (\beta_0 + \beta_2\delta_0) + (\beta_1 + \beta_2\delta_1)x_1 + (\beta_2 v + u)$$

If y is only regressed on x_1 this will be the estimated intercept

If y is only regressed on x_1, this will be the estimated slope on x_1

error term

- Conclusion: All estimated coefficients will be biased

The expected value of the OLS Estimators

- **Omitting relevant variables: the simple case (cont.)**

Using

$$\tilde{\beta}_1 = \frac{\sum_{i=1}^{n}(x_{i1} - \bar{x}_1)y_i}{\sum_{i=1}^{n}(x_{i1} - \bar{x}_1)^2}$$

one can derive:

$$E(\tilde{\beta}_1) = \beta_1 + \beta_2 \frac{\sum_{i=1}^{n}(x_{i1} - \bar{x}_1)x_{i2}}{\sum_{i=1}^{n}(x_{i1} - \bar{x}_1)^2}$$

or

$$E(\tilde{\beta}_1) = \beta_1 + \beta_2 \tilde{\delta}_1$$

where $\tilde{\delta}_1$ is the slope coefficient in the simple regression: x_{i2} on x_{i1}, $i = 1, \ldots, n$

The expected value of the OLS Estimators

- Therefore, the bias in $\tilde{\beta}_1$ is:



$$Bias(\tilde{\beta}_1) = \beta_2 \tilde{\delta}_1$$

Called the **omitted variable bias**

- The simple regression estimator is unbiased in two cases:
  - $\beta_2 = 0$. But this means that x_2 does not appear in the population model.
  - $Corr(x_{i1}, x_{i2}) = 0$. Then the simple and multiple regression estimators are identical because $\tilde{\delta}_1 = 0$.
- Summary of bias in the simple regression estimator of $\tilde{\beta}_1$.

	$Corr(x_1, x_2) > 0$	$Corr(x_1, x_2) < 0$
$\beta_2 > 0$	Positive Bias	Negative Bias
$\beta_2 < 0$	Negative Bias	Positive Bias

The expected value of the OLS Estimators

- Example: Omitting ability in a wage equation

$$wage = \beta_0 + \beta_1 educ + \beta_2 abil + u$$

$$abil = \delta_0 + \delta_1 educ + v$$

Will both be positive

$$wage = (\beta_0 + \beta_2\delta_0) + (\beta_1 + \beta_2\delta_1) educ + (\beta_2 v + u)$$

The return to education β_1 will be overestimated because $\beta_2\delta_1 > 0$ It will look as if people with many years of education earn very high wages, but this is partly due to the fact that people with more education are also more able on average.

- When is there no omitted variable bias?
  - If the omitted variable is irrelevant or uncorrelated

The expected value of the OLS Estimators

- Omitted variable bias: more general cases

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_3 + u$$ ← True model (contains x_1, x_2, and x_3)

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + w$$ ← Estimated model (x_3 is omitted)

  - No general statements possible about direction of bias
  - Analysis as in simple case if one regressor uncorrelated with others

- Example: Omitting ability in a wage equation

$$wage = \beta_0 + \beta_1 educ + \beta_2 exper + \beta_3 abil + u$$

If exper is approximately uncorrelated with educ and abil, then the direction of the omitted variable bias can be as analyzed in the simple two variable case.

Example

Empirical problem: Class size and educational output

$$\widehat{\text{TestScore}} = \underset{(10.36)}{698.9} - \underset{(0.52)}{2.28} \times \text{STR}, \; R^2 = 0.05, SER = 18.58$$

Is this a credible estimate of the causal effect on test scores of a change in the student-teacher ratio?

- No: there are omitted confounding factors (family income, English ability) that may bias the OLS estimator: STR could be "picking up" the effect of these confounding factors

Example

With the tool of multiple regression in hand, we may estimate the causal effect of STR by adding important factors.

School characteristics:

- Student-teacher ratio
- Teacher quality
- Computers (non-teaching resources) per student
- Measures of curriculum design ...

Student characteristics:

- English proficiency
- Availability of extracurricular enrichment
- Home learning environment
- Parent's education level...

Example

To do this, we need to think about what variables to include – we should do this before we actually sit down at the computer. This entails thinking beforehand about your model specification

A general approach to model specification (variable selection):

- Specify a "base" or "benchmark" model
- Specify a range of plausible alternative models, which include additional candidate variables
- Does a candidate variable change the coefficient of interest?
- Is a candidate variable statistically significant?
- Use judgment and economic theory ...

Example



It is easy to fall into the trap of maximizing the R^2 and $\bar{R}^2$. Note that our real objective is to obtain a consistent estimator of the class size effect.

- A high R^2 (or $\bar{R}^2$) means that the regressors explain the variation in Y well
- A high R^2 (or $\bar{R}^2$) does not mean that you have eliminated omitted variable bias
- A high R^2 (or $\bar{R}^2$) does not mean that you have an unbiased estimator of β_1
- A high R^2 (or $\bar{R}^2$) does not mean that the included variables are statistically significant – this must be determined using hypotheses tests

TABLE 5.2 Results of Regressions of Test Scores on the Student-Teacher Ratio and Student Characteristic Control Variables Using California Elementary School Districts

Dependent variable: Average test score in the district.

Regressor	(1)	(2)	(3)	(4)	(5)
Student-teacher ratio (X_1)	−2.28**	−1.10*	−1.00**	−1.31**	−1.01**
	(0.52)	(0.43)	(0.27)	(0.34)	(0.27)
Percent English learners (X_2)		−0.650**	−0.122**	−0.488**	−0.130**
		(0.031)	(0.033)	(0.030)	(0.036)
Percent eligible for subsidized lunch (X_3)			−0.547**		−0.529**
			(0.024)		(0.038)
Percent on public income assistance (X_4)				−0.790**	0.048
				(0.068)	(0.059)
Intercept	698.9**	686.0**	700.2**	698.0**	700.4**
	(10.4)	(8.7)	(5.6)	(6.9)	(5.5)
Summary Statistics					
SER	18.58	14.46	9.08	11.65	9.08
$\overline{R}^2$	0.049	0.424	0.773	0.626	0.773
n	420.0	420.0	420.0	420.0	420.0

These regressions were estimated using the data on K–8 school districts in California, described in Appendix 4.1. Standard errors are given in parentheses under coefficients. The individual coefficient is statistically significant at the *5% level or **1% significance level using a two-sided test.

Summary

- Multiple regression allows you to estimate the causal effect of a regressor on Y by adding control variables
- If you can measure a variable, you can avoid omitted variable bias from that variable by including it
- There is no simple recipe for deciding which variables belong in a regression – you must exercise judgment
- One approach is to specify a base model (relying on *a priori* reasoning), and then explore the sensitivity of the key estimate(s) in alternative specifications

The variance of the OLS Estimators

- Standard assumptions for the multiple regression model (cont.)
- Assumption MLR.5 (Homoskedasticity)

$$Var(u_i|x_{i1}, x_{i2}, \ldots, x_{ik}) = \sigma^2$$

The value of the explanatory variables must contain no information about the variance of the unobserved factors

- Example: Wage equation

$$Var(u_i|educ_i, exper_i, tenure_i) = \sigma^2$$

This assumption may also be hard to justify in many cases

- Short hand notation

$$Var(u_i|\mathbf{x}_i) = \sigma^2 \quad \text{with} \quad \mathbf{x}_i = (x_{i1}, x_{i2}, \ldots, x_{ik})$$

All explanatory variables are collected in a random vector

The variance of the OLS Estimators

Gauss Markov assumptions in MLR:

- MLR.1: $y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + ... + \beta_k x_k + u$
- MLR.2: random sampling from the population
- MLR.3: no perfect collinearity in the sample
- MLR.4: $E(u|x_1, ..., x_k) = E(u) = 0$ (exogenous explanatory variables)
- MLR.5: $Var(u|x_1, ..., x_k) = Var(u) = \sigma^2$ (homoskedasticity)

The variance of the OLS Estimators

- Assumptions MLR.1 through MLR.5 are called the **Gauss Markov assumptions.**

- Assumptions MLR.1 and MLR.4 imply

$$E(y|x_1, x_2, ..., x_k) = \beta_0 + \beta_1 x_1 + ... + \beta_k x_k$$

- and when we add MLR.5,

$$Var(y|x_1, x_2, ..., x_k) = Var(u|x_1, x_2, ..., x_k) = \sigma^2$$

The variance of the OLS Estimators

- Theorem 3.2 (Sampling variances of the OLS slope estimators)

  Under assumptions MLR.1 – MLR.5:

Variance of the error term

$$Var(\hat{\beta}_j) = \frac{\sigma^2}{SST_j(1 - R_j^2)}, \quad j = 1, \ldots, k$$

Total sample variation in explanatory variable x_j:

$$\sum_{i=1}^{n} (x_{ij} - \bar{x}_j)^2$$

R-squared from a regression of explanatory variable x_j on all other independent variables (including a constant)

The variance of the OLS Estimators

i = observation

j = variable

- **Variances of the OLS slope esimators (Chapter 3 Appendix)**

  In the model with *k* independent variables, we can write $\hat{\beta}_j$ as:

$$\hat{\beta}_j = \beta_j + \frac{\sum_{i=1}^{n} \hat{r}_{ij} u_i}{\sum_{i=1}^{n} \hat{r}_{ij}^2}$$

where $\hat{r}_{ij}$ is the residual from a regression of x_j on $x_1, \ldots, x_{j-1}, x_{j+1}, \ldots, x_k$

$$Var(\hat{\beta}_j) = Var\left(\frac{\sum_{i=1}^{n} \hat{r}_{ij} u_i}{\sum_{i=1}^{n} \hat{r}_{ij}^2}\right) = \frac{\sum_{i=1}^{n} \hat{r}_{ij}^2 Var(u_i)}{(\sum_{i=1}^{n} \hat{r}_{ij}^2)^2}$$

$$= \frac{\sigma^2}{\sum_{i=1}^{n} \hat{r}_{ij}^2} = \frac{\sigma^2}{SSR_j} = \frac{\sigma^2}{SST_j(1 - R_j^2)}$$

The variance of the OLS Estimators

- **Components of OLS Variances:**
- **1) The error variance**
  - A high error variance increases the sampling variance because there is more “noise” in the equation
  - A large error variance necessarily makes estimates imprecise
  - The error variance does not decrease with sample size
- **2) The total sample variation in the explanatory variable**
  - More sample variation leads to more precise estimates
  - Total sample variation automatically increases with the sample size
  - Increasing the sample size is thus a way to get more precise estimates

The variance of the OLS Estimators

- **3) Linear relationships among the independent variables**

  R_j^2 : Regress x_j on all other independent variables (including a constant)

  The R-squared of this regression measures how linearly related x_j is to the other explanatory variables.

  - Sampling variance of $\hat{\beta}_j$ will be the higher the better explanatory variable x_j can be linearly explained by other independent variables
  - The problem of almost linearly dependent explanatory variables is called **multicollinearity** (i.e. $R_j \to 1$ for some j)
  - Multicollinearity can cause problems for getting a imprecise estimate

$$R_j^2 \to 1,\ Var(\hat{\beta}_j) \to \infty$$

"Rule of thumb":
$R_j^2 > 0.9$

The variance of the OLS Estimators

- **Example: Factors Affecting NBA Salaries**

  Data: NBASAL.DTA

Annual salary, thousands $

Averge minutes per game

$$\widehat{lwage} = \underset{(.1101)}{5.56} + \underset{(.004)}{.058}\, avgmin$$

Averge points per game

$$\widehat{lwage} = \underset{(.0846)}{6.01} + \underset{(.0072)}{.093}\, points$$

$$\widehat{lwage} = \underset{(.116)}{5.66} + \underset{(.0091)}{.039}\, avgmin + \underset{(.015)}{.035}\, points$$

Both the stardard errors of $avgmin$ and $points$ increase dramatically in the MLR

⇒ 高 correlation

The variance of the OLS Estimators

- An example for multicollinearity

Average standardized test score of school

Expenditures for teachers

Expenditures for instructional materials

Other expenditures

$$avgscore = \beta_0 + \beta_1 teachexp + \beta_2 matexp + \beta_3 othexp + \ldots$$

The different expenditure categories will be strongly correlated because if a school has a lot of resources it will spend a lot on everything.

It will be hard to estimate the differential effects of different expenditure categories because all expenditures are either high or low. For precise estimates of the differential effects, one would need information about situations where expenditure categories change differentially.

As a consequence, sampling variance of the estimated effects will be large.

The variance of the OLS Estimators

- **Discussion of the multicollinearity problem**
  - In the previous example, it would probably be better to lump all expenditure categories together because effects cannot be disentangled
  - In other cases, dropping some independent variables may reduce multicollinearity (but this may lead to omitted variable bias)
  - Only the sampling variance of the variables involved in multicollinearity will be inflated; the estimates of other effects may be very precise, for example:

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_3 + u,$$

where β_1 is the coefficient of interest. In fact, assume x_2 and x_3 act as controls so that we hope to get a good ceteris paribus estimate of x_1. The key is that the correlation between x_2 and x_3 has nothing to do with $Var(\hat{\beta}_1)$, it is only the correlation of x_1 with (x_2, x_3) that matters.

The variance of the OLS Estimators

- **Discussion of the multicollinearity problem (cont.)**
  - Note that multicollinearity does not violate any of the Gauss-Markov assumptions, including MLR.3.
  - R_j^2 "close" to one is called the " problem" of multicollinearity. Unfortunately, we cannot define what we mean by "close" that is relevant for all situations.
- Detecting multicollinearity through **variance inflation factors**

$$VIF_j = 1/(1 - R_j^2)$$

$$Var(\hat{\beta}_j) = \frac{\sigma^2}{SST_j} \cdot VIF_j$$

$$Var(\hat{\beta}_j) = Var\left(\frac{\sum_{i=1}^n \hat{r}_{ij} u_i}{\sum_{i=1}^n \hat{r}_{ij}^2}\right) = \frac{\sum_{i=1}^n \hat{r}_{ij}^2 Var(u_i)}{(\sum_{i=1}^n \hat{r}_{ij}^2)^2}$$

$$= \frac{\sigma^2}{\sum_{i=1}^n \hat{r}_{ij}^2} = \frac{\sigma^2}{SSR_j} = \frac{\sigma^2}{SST_j(1 - R_j^2)}$$

- the VIF_j tells us how many times larger the variance is than if we had the "ideal" case of no correlation of x_j with the other explanatory variables.
- As an (arbitrary) rule of thumb, the variance inflation factor should not be larger than 10.
- There are ways to offset the variance inflation factor. Ultimately, it is $Var(\hat{\beta}_j)$ that is important.

The variance of the OLS Estimators

- **Theorem 3.1 (Unbiasedness of OLS)**

$$MLR.1-MLR.4 \Rightarrow E(\hat{\beta}_j) = \beta_j, \quad j = 0, 1, \ldots, k$$

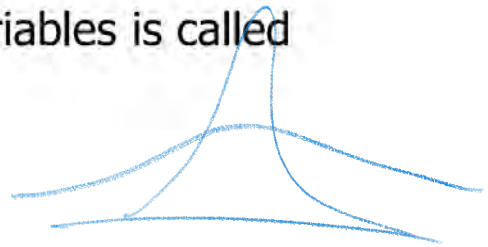
Leaving a variable out when it should be included in multiple regression is called **underspecifying the model**, which leads to **omitted variable bias.**

- **Theorem 3.2 (Sampling variances of OLS slope estimators)**

$$MLR.1-MLR.5 \Rightarrow Var(\hat{\beta}_j) = \frac{\sigma^2}{SST_j(1-R_j^2)}, \quad j = 1, \ldots, k$$

The problem of almost linearly dependent explanatory variables is called **multicollinearity** (i.e. $R_j \to 1$ for some j)

will not bias, only not precise



The variance of the OLS Estimators

- Variances in misspecified models
  - The choice of whether to include a particular variable in a regression can be made by analyzing the tradeoff between bias and variance

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + u$$ ← True population model

$$\hat{y} = \hat{\beta}_0 + \hat{\beta}_1 x_1 + \hat{\beta}_2 x_2$$ ← Estimated model 1

$$\tilde{y} = \tilde{\beta}_0 + \tilde{\beta}_1 x_1$$ ← Estimated model 2

  - It might be the case that the likely omitted variable bias in the misspecified model 2 is overcompensated by a smaller variance

The variance of the OLS Estimators

- Variances in misspecified models (cont.)

$$Var(\hat{\beta}_1) = \sigma^2 / \left[SST_1(1 - R_1^2)\right]$$

$$Var(\tilde{\beta}_1) = \sigma^2 / SST_1$$

Conditional on x_1 and x_2, the variance in model 2 is always smaller than that in model 1

- Case 1:

Conclusion: Do not include irrelevant regressors

$$\beta_2 = 0 \Rightarrow E(\hat{\beta}_1) = \beta_1, E(\tilde{\beta}_1) = \beta_1, Var(\tilde{\beta}_1) < Var(\hat{\beta}_1)$$

- Case 2:

Trade off bias and variance; Caution: bias will not vanish even in large samples

$$\beta_2 \neq 0 \Rightarrow E(\hat{\beta}_1) = \beta_1, E(\tilde{\beta}_1) \neq \beta_1, Var(\tilde{\beta}_1) < Var(\hat{\beta}_1)$$

The variance of the OLS Estimators

- Estimating the error variance

$$\hat{\sigma}^2 = \left(\sum_{i=1}^{n} \hat{u}_i^2 \right) / [n - k - 1]$$

An unbiased estimate of the error variance can be obtained by substracting the number of estimated regression coefficients from the number of observations. The number of observations minus the number of estimated parameters is also called the degrees of freedom. The n estimated squared residuals in the sum are not completely independent but related through the k+1 equations that define the first order conditions of the minimization problem.

$$MLR.1 - MLR.5 \quad \Rightarrow \quad E(\hat{\sigma}^2) = \sigma^2$$

- Theorem 3.3 (Unbiased estimator of the error variance)

The variance of the OLS Estimators

- Estimation of the sampling variances of the OLS estimators

The true sampling variation of the estimated β_j

$$sd(\hat{\beta}_j) = \sqrt{Var(\hat{\beta}_j)} = \sqrt{\sigma^2 / \left[SST_j(1 - R_j^2)\right]}$$

Plug in $\hat{\sigma}^2$ for the unknown σ^2

The estimated sampling variation of the estimated β_j

$$se(\hat{\beta}_j) = \sqrt{\widehat{Var}(\hat{\beta}_j)} = \sqrt{\hat{\sigma}^2 / \left[SST_j(1 - R_j^2)\right]}$$

- Note that these formulas are only valid under assumptions MLR.1-MLR.5 (in particular, there has to be homoskedasticity)